DIET: Machine Unlearning on a Data-Diet

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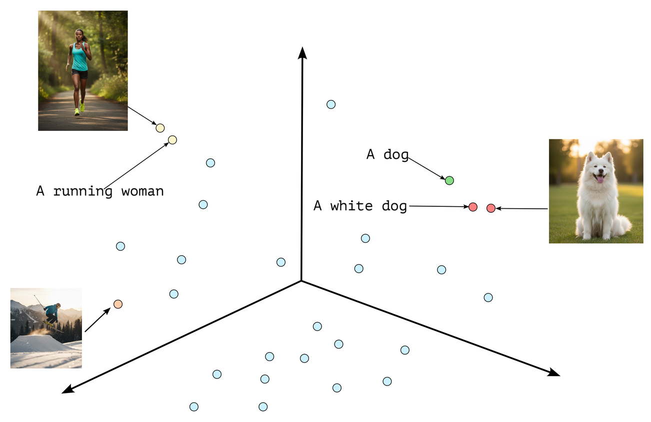
SCAN ME!

Why Retain-Free for VLMs?

- **Challenge:** Machine Unlearning (MUL) is crucial for data governance and privacy compliance.
- **Limitations:**
  - Rely on **Retain Data** to reduce catastrophic forgetting.
  - This is costly, risks privacy breaches, and struggles to scale.
- **VLM-Specific Challenge:** Unlearning in **Vision-Language Models (VLMs)** (e.g., CLIP) is extra challenging due to *highly entangled multimodal embeddings*.



Language as Retain-Free Solution for VLMs. Use text prototypes as a retain-free mechanism to **guide unlearning** - no need for image–text retain pairs.

Key Contributions:

- We propose DIET: A **principled, retain-data-free** unlearning method for VLMs.
- Unlearning by achieving **asymptotically negligible influence** of forget data via **hyperbolic geometry** (Poincaré Ball Model) **effectively PUSHING the concept to infinity.**
- Achieves **comparable utility to data-heavy methods** while being **retain-free**, with **117.5% utility improvement** over best retain-free baselines.

The Geometry of Forgetfulness

Hyperbolic Space for Retain-Free Unlearning:

- **Goal:** To "forget", the model must push unwanted embeddings far away-ideally to **infinity**, where they have no semantic influence.
- **Euclidean** ($\mathbb{R}^n$): "Infinity" is ill-defined; no tractable distance measure to infinity.
- **Poincaré Ball Model** ($\mathbb{B}^n$): The canonical model of Hyperbolic space ($\mathbb{H}^n$), $\mathbb{B}^n = \{\mathbf{z} \in \mathbb{R}^n : \|\mathbf{z}\| < 1\}$ where the unit sphere ($\partial\mathbb{B}$) serves as the **ideal boundary, representing hyperbolic infinity**.

Busemann Function ($\delta_{\mathbf{p}}(\mathbf{z})$)**:** Measures the hyperbolic distance from $\mathbf{z} \in \mathbb{B}^n$ to points at the boundary $\mathbf{p} \in \partial\mathbb{B}^n$.

$$\delta_{\mathbf{p}}(\mathbf{z}) = \lim_{t\to\infty} \left(d_{\mathbb{B}}(\gamma_{\mathbf{p}}(t), \mathbf{z}) - t\right) \quad (1)$$

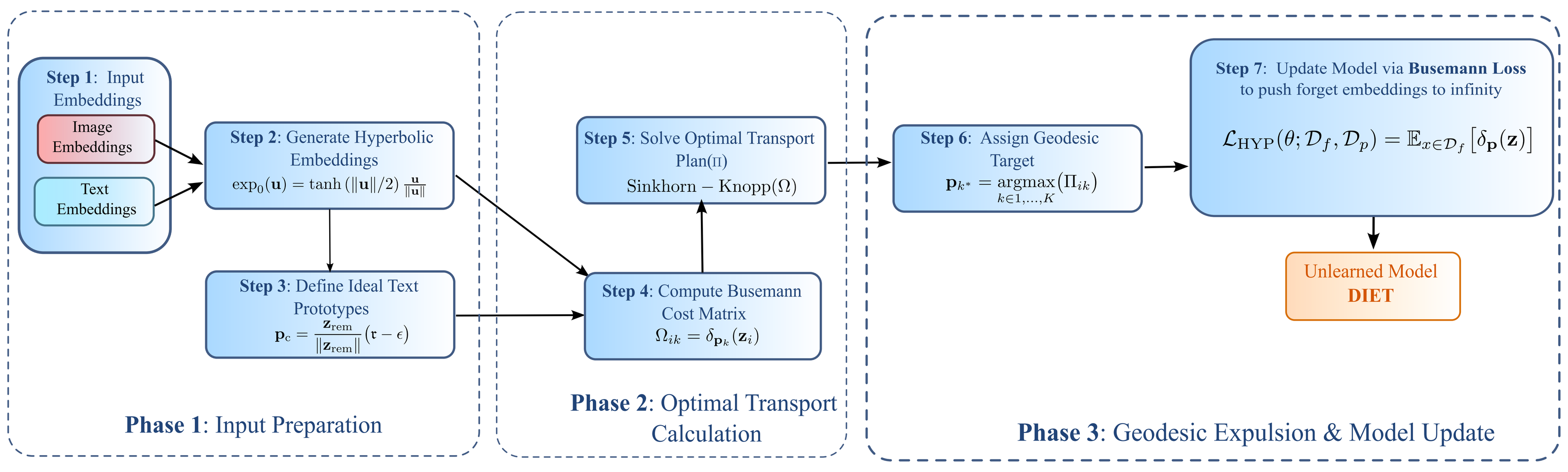
The **Busemann distance** tells us, **at the infinite boundary**, how much "excess" **hyperbolic distance** remains between an **interior point** $\in \mathbb{B}^n$ and its **ideal path to infinity**, after normalizing for the **path length already traversed**.

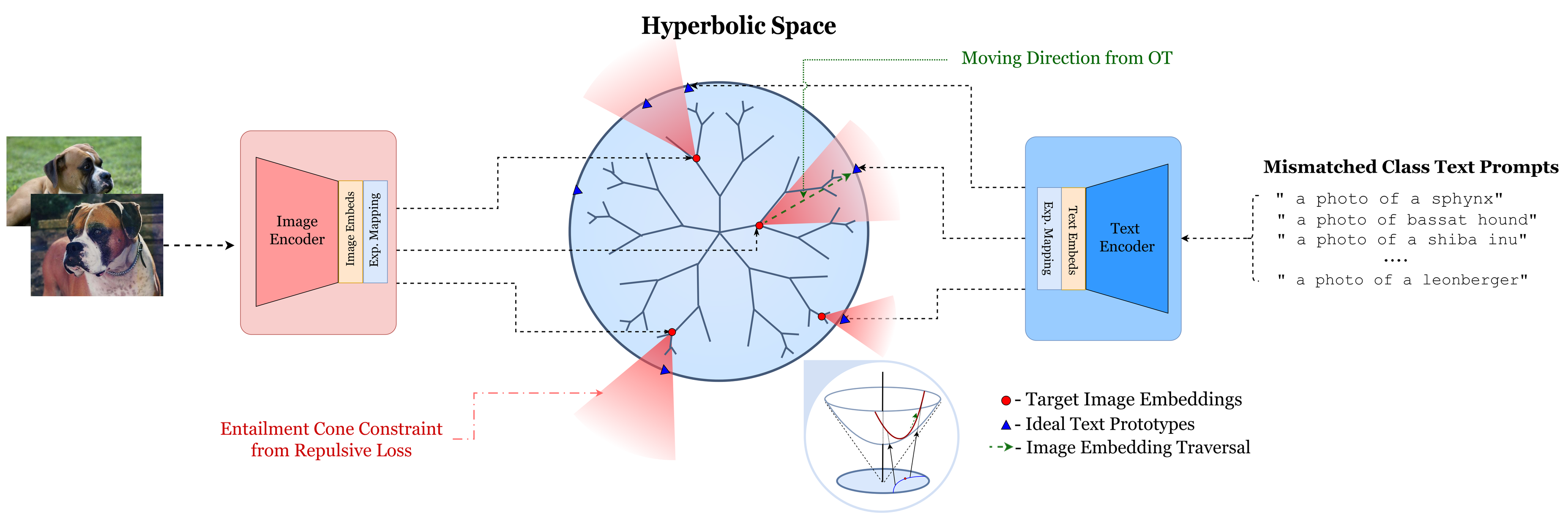
Unlearning Action: Minimizing $\delta_p(\mathbf{z})$ forcefully drives the forget embedding $\mathbf{z}$ along a geodesic path **directly towards p at infinity** ($\partial\mathbb{B}$).

The DIET Framework

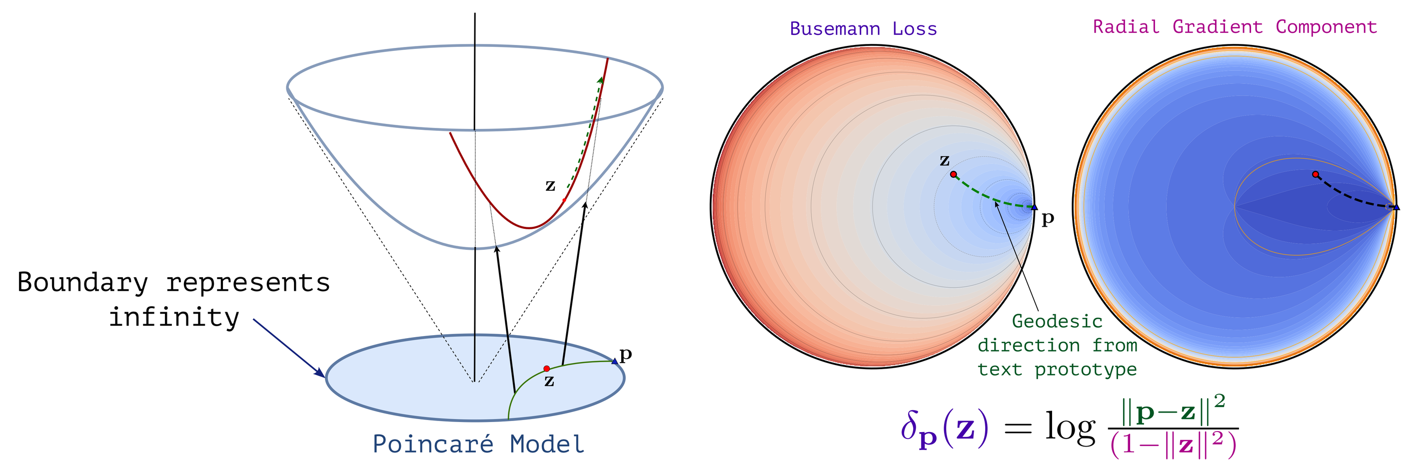
Semantic Erasure As embeddings approach the boundary, **their influence becomes infinitely distant** from interior points, effectively **erasing semantic contribution**.

The Road to Infinity:





Busemann Loss:



Repulsive Loss: Surgical Precision for Unlearning
Ensures the geodesic paths for unlearning remain within the entailment cone.
This guarantees **surgical precision**, preventing collateral damage to retained knowledge.

Guiding the Forgetting:

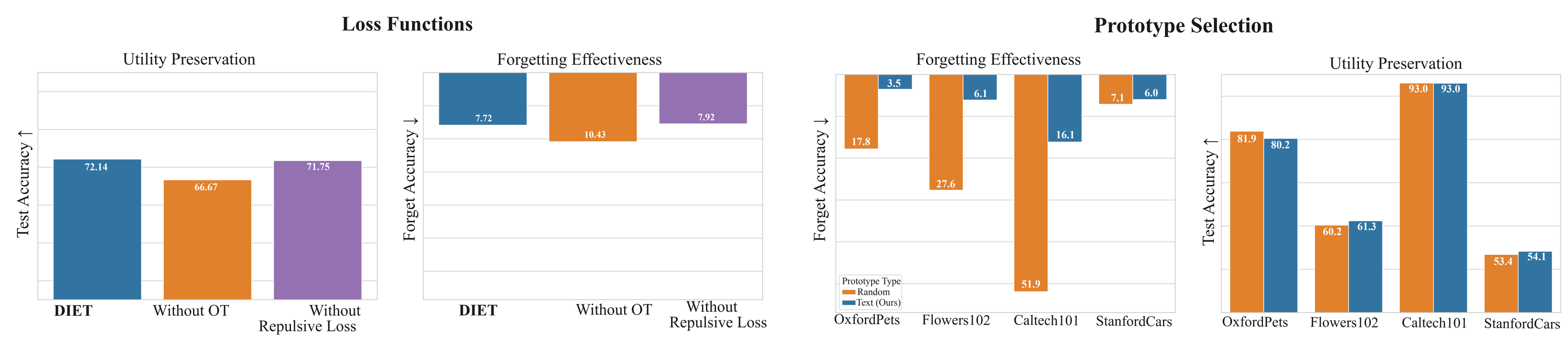
- *Target Direction (Language):* Text prototypes (our retain-free signal) **define the ideal boundary point** p (infinity) for the concept to be forgotten.
- *Optimal Transport (OT):* Finds **an optimal, adaptive assignment** between forget image embeddings and the set of mismatched text prototypes. This initiates the push.
- *Busemann Function:* Actively steers forget embeddings along the defined **hyperbolic geodesics** towards the boundary, based on the OT assignment.

Results

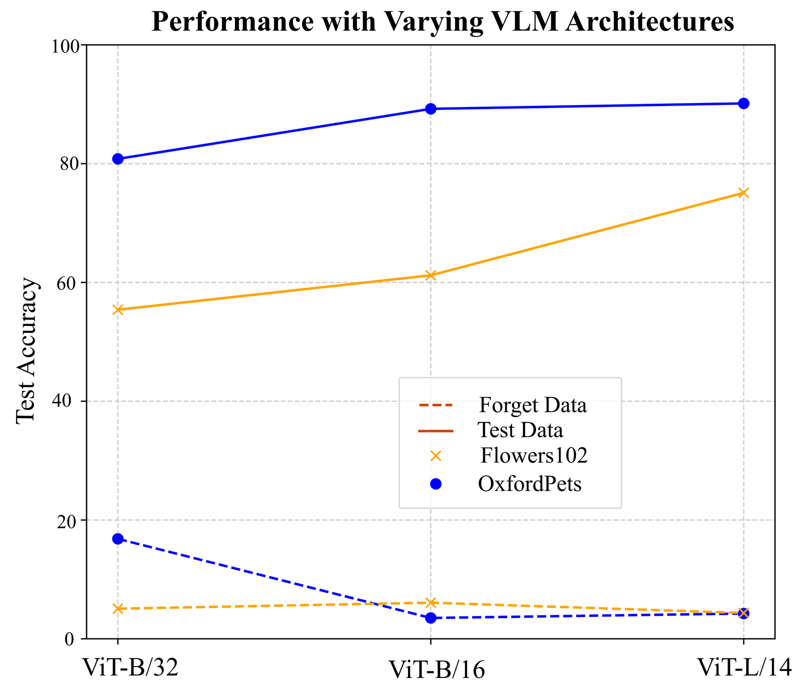
Quantitative Results: DIET achieves **8.06% average forget accuracy** and **69.04% utility preservation** using only minimal forget samples (16 per concept).

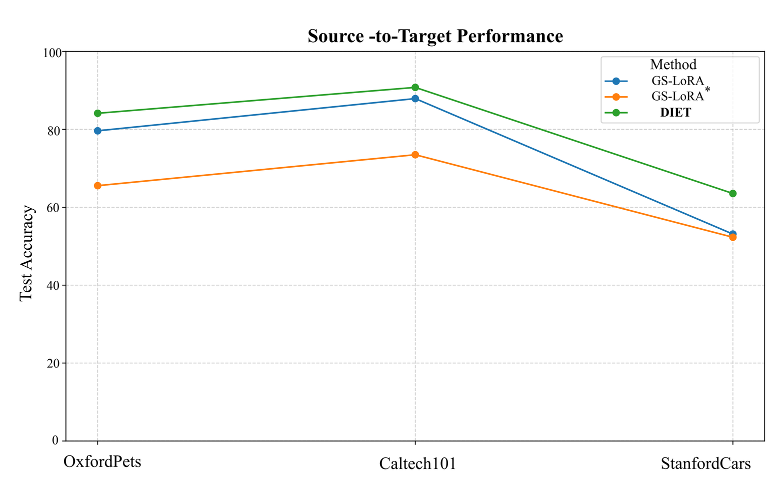
	Method	D_r	Flowers102		Pets		Cars		Food101		Average	
			$\mathcal{D}_f$ (%)	$\mathcal{D}_t$ (%)	$\mathcal{D}_f$ (%)	$\mathcal{D}_t$ (%)	$\mathcal{D}_f$ (%)	$\mathcal{D}_t$ (%)	$\mathcal{D}_f$ (%)	$\mathcal{D}_t$ (%)	$\mathcal{D}_f$ (%)	$\mathcal{D}_t$ (%)
Pretrained	Zero-shot	-	92.61	70.48	83.25	89.12	53.43	65.58	87.20	86.12	79.12	77.82
Weights updated	FT	✓	56.58	82.07	62.74	88.34	35.12	60.32	70.68	79.95	56.28	77.67
	GA	✓	86.55	70.81	74.75	88.52	46.92	65.18	84.68	84.61	73.23	77.28
	SHs	✓	0.38	28.28	0.00	3.53	0.00	0.73	7.00	15.39	1.85	11.98
	SalUn	✓	21.71	43.59	14.81	59.52	28.10	66.21	55.80	84.47	30.11	63.45
	SalUn*	✗	24.44	56.19	26.00	55.04	7.08	22.85	36.16	40.05	23.42	43.53
LoRA based	GA	✗	0.00	17.71	0.00	45.25	0.00	7.15	0.00	56.86	0.00	31.74
	GS-LoRA	✓	**0.00**	**68.15**	**0.25**	**84.33**	**0.00**	**54.60**	**0.28**	**79.95**	**0.13**	**71.76**
	GS-LoRA*	✗	0.00	1.66	0.00	2.89	0.00	0.70	0.00	1.67	0.00	1.73
	DIET (Ours)	✗	6.08	61.20	3.51	80.23	5.97	54.14	16.68	80.59	8.06	69.04

* indicates performance without using a retain dataset.



Scalable and Generalizable Unlearning:





Visualizing the Forgetting Effect: